Data collection and preprocessing

- Electrictity consumption data (Umeå Energi)
- Weather data (SMHI)
- Calendar data (Dagsmart API)

Datasets

Data resampling

- Downscaling weather data from hourly to monthly data

Feature configuration

- Constructing lag features for historical consumption
- Household clustering using K-Means
- THI and WCT calculation
- Construct num. holidays and num. days features

Model construction and evaluation

RF, GBM, XGBoost, LightGBM, CatBoost

Train-test data split

- 69% train data, 31% test data

Hyperparameter optimization (grid search)

Ensemble model construction

- Using best hyperparameters

Performance evaluation

- MAPE, RMSE, MAE metrics
- Comparing ensemble models
- Comparing with baseline linear regression model

Model explanation

Using SHAP method

Explainable AI analysis of forecasts

- SHAP summary (beeswarm) plot
- Feature importance